

Development of a Novel Sensor for Measuring Electron Energisation in Space Plasmas

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Scientific Motivation

Why Electron Energisation?

Electron energisation is a ubiquitous process limited not just to the heliosphere but throughout the wider universe. Due to the collisionless nature of most plasmas in space, electrons gain energy through interaction with electric fields rather than Coulomb collisions. Measuring these interactions requires high cadence measurements of electron velocity distribution functions with velocity-space resolution sufficient to distinguish energy transfer, resonance and kinetic-scale structure.

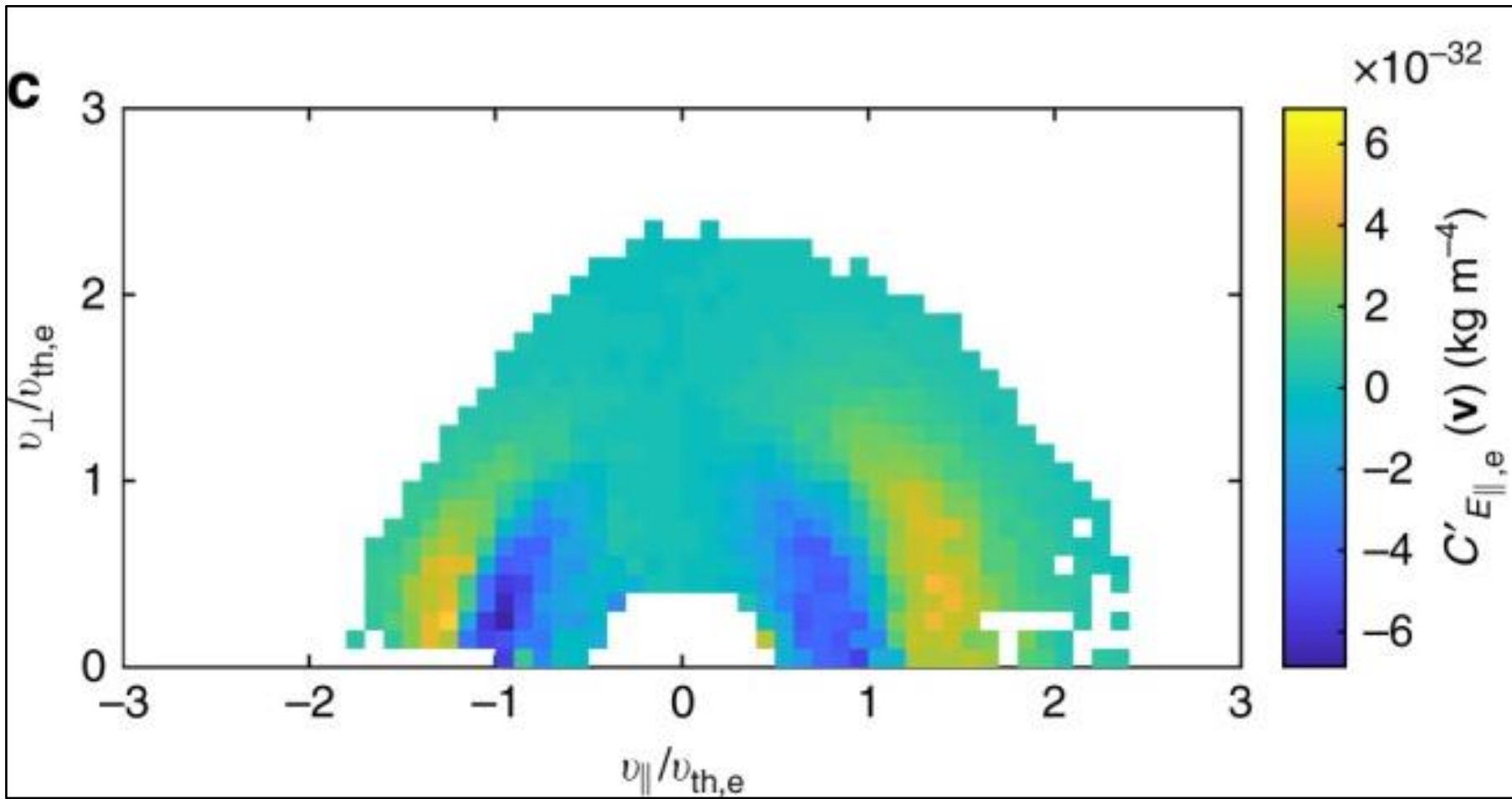
Electrons are energised via wave-particle interactions such as:

- Landau Damping
- Cyclotron Damping
- Magnetic Pumping
- Transit Time Damping

Instrument Requirements:

- Electron VDF cadence $\leq 1\text{--}5\text{ms}$
- Velocity-space resolution $\Delta v/v \lesssim 10\%$
- Velocity coverage up to $\sim 11,000\text{km/s}$
- Sufficient count statistics for reliable VDF reconstruction

- Electron energisation produces structured, non-thermal VDFs
- These features evolve on short timescales
- This motivates requirements on cadence, angular coverage, and energy sampling



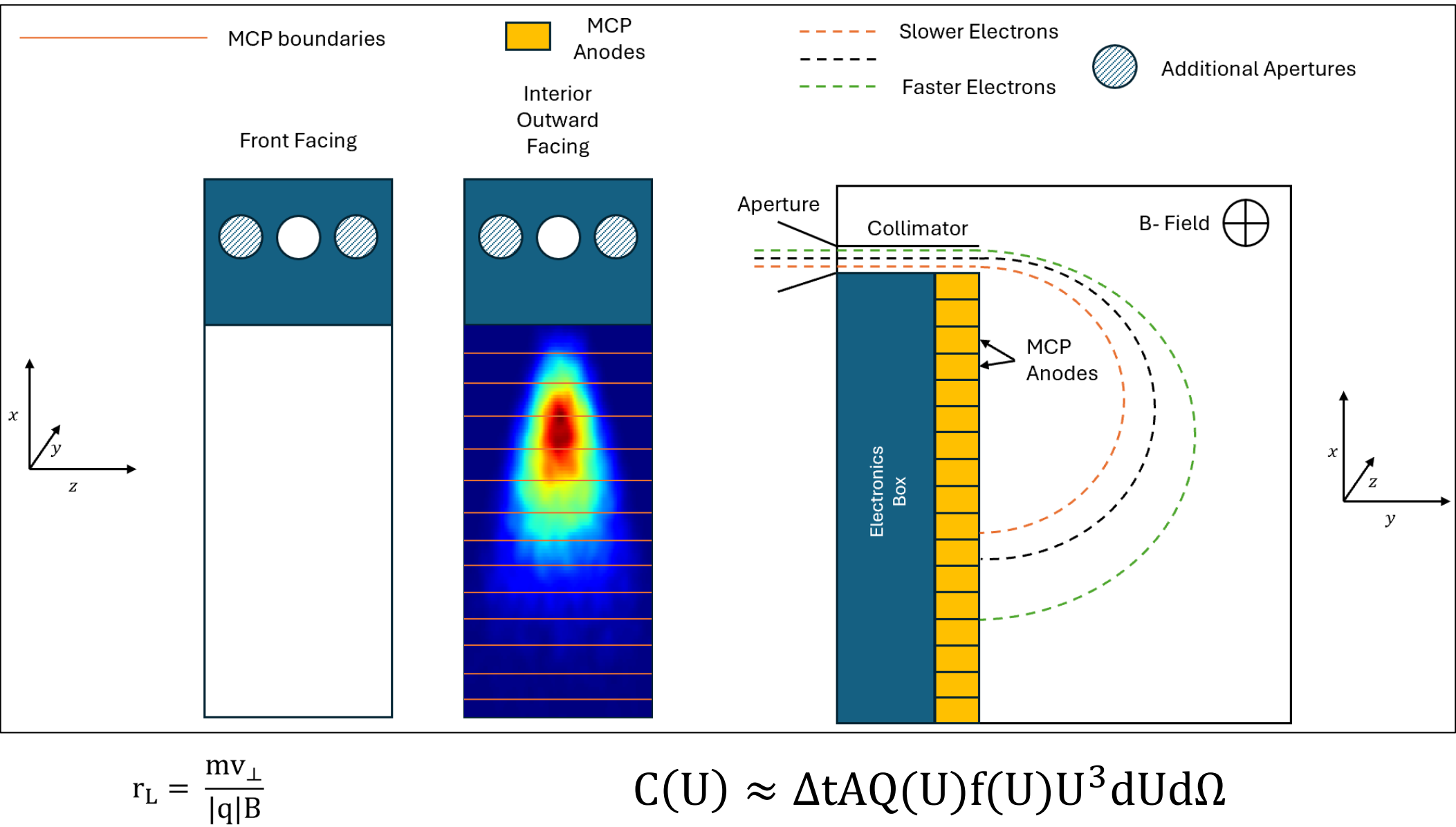
Electron Velocity Distribution showing field-particle energy transfer (Electron Landau Damping).

Velocity-space map of $C'_{E||,e}(\mathbf{v})$ as a function of normalized parallel and perpendicular electron velocity. Positive and negative regions indicate where electrons gain or lose energy through parallel electric-field interactions, revealing signatures of electron energisation in phase space [1].

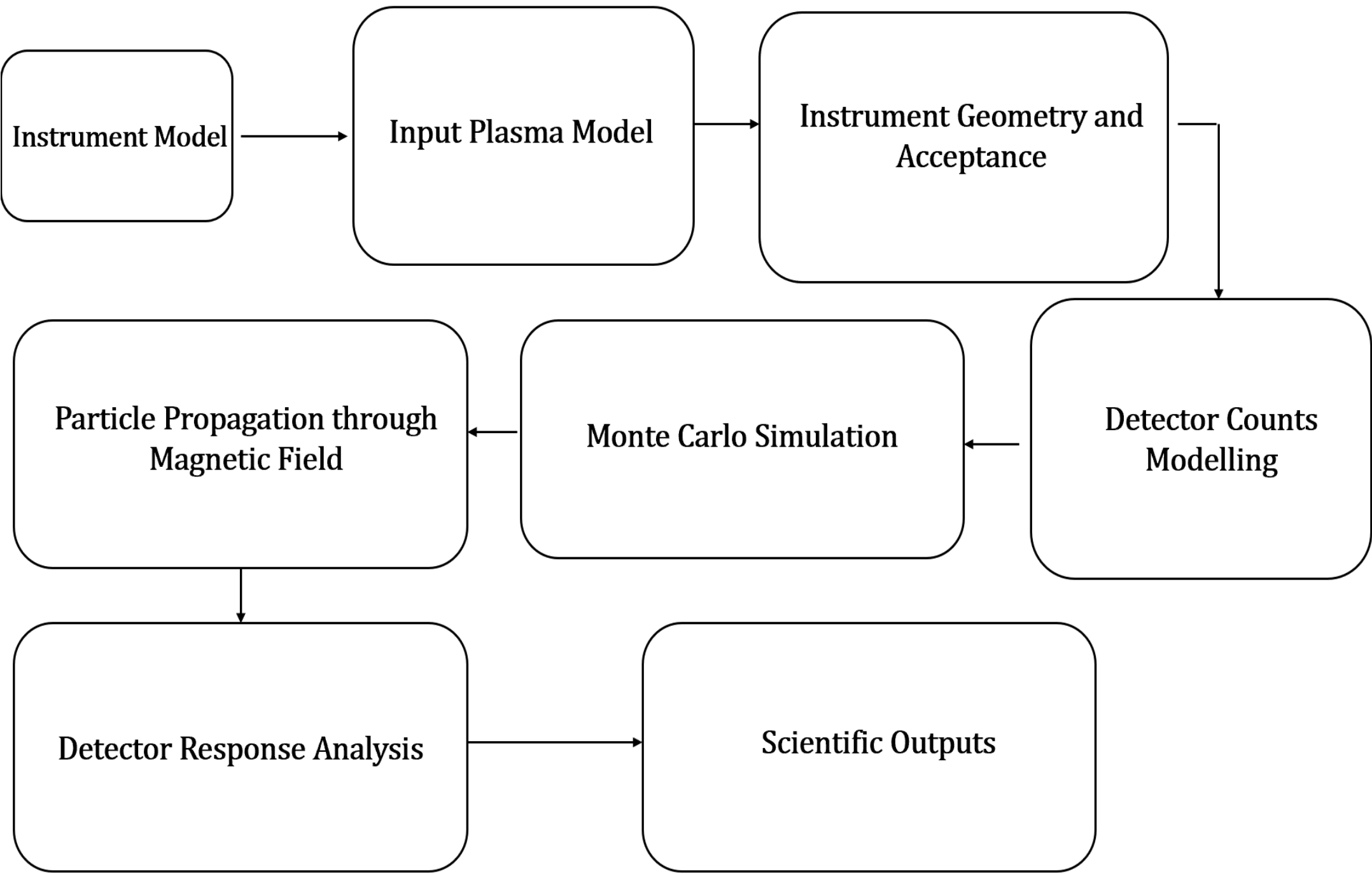
Instrument Concept

Magnetic Deflection Concept

- Electrons enter through collimated apertures
- Internal magnetic field bends trajectories proportional to velocity
- Impact position on MCP/anode array maps to electron velocity
- Multiple apertures increase geometric factor and count rate
- No voltage sweeping required $\rightarrow$ potential millisecond cadence
- Compact, low-SWaP design for multi-spacecraft missions



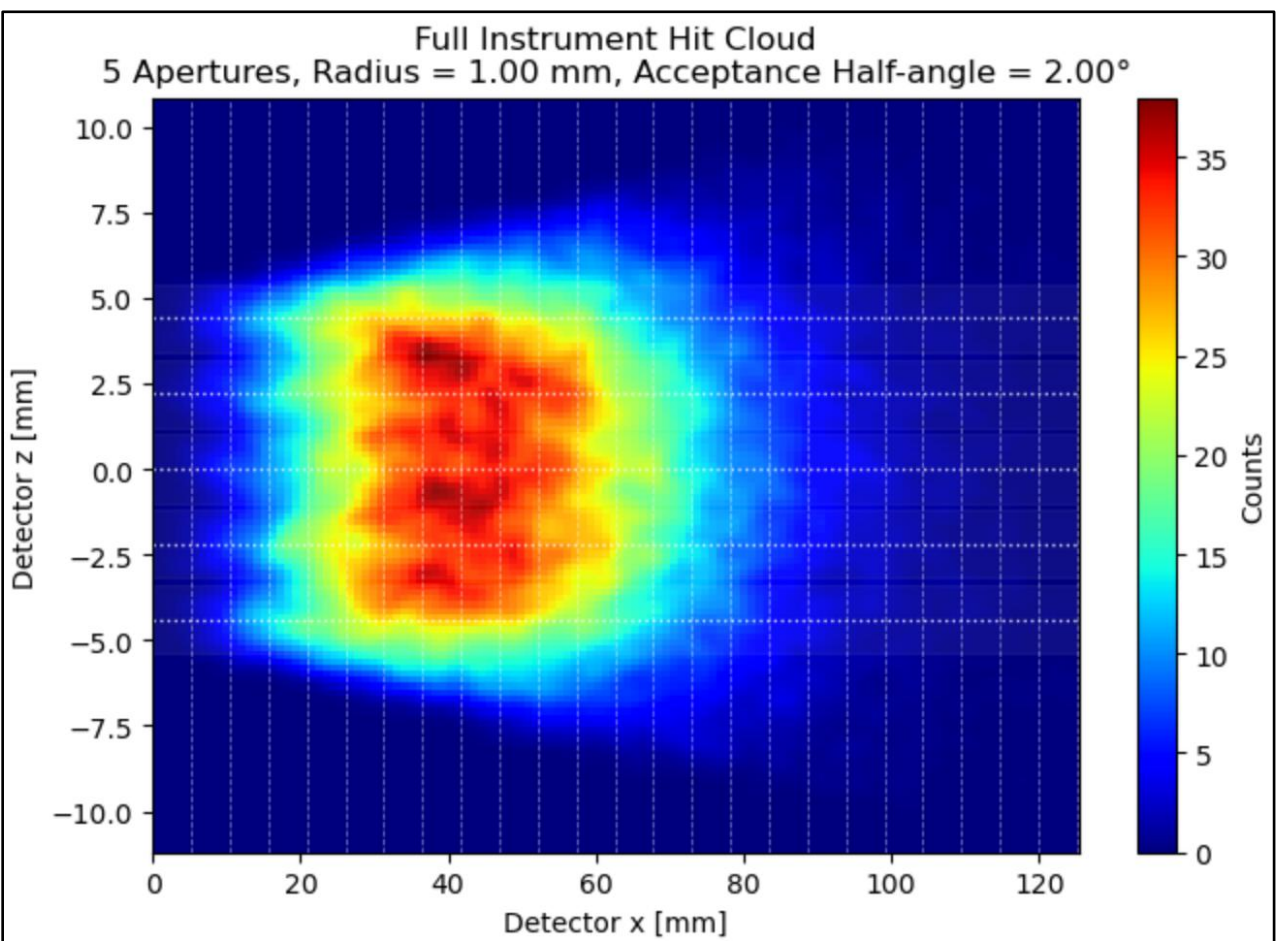
Simulation Framework Designed for Instrument Performance Study



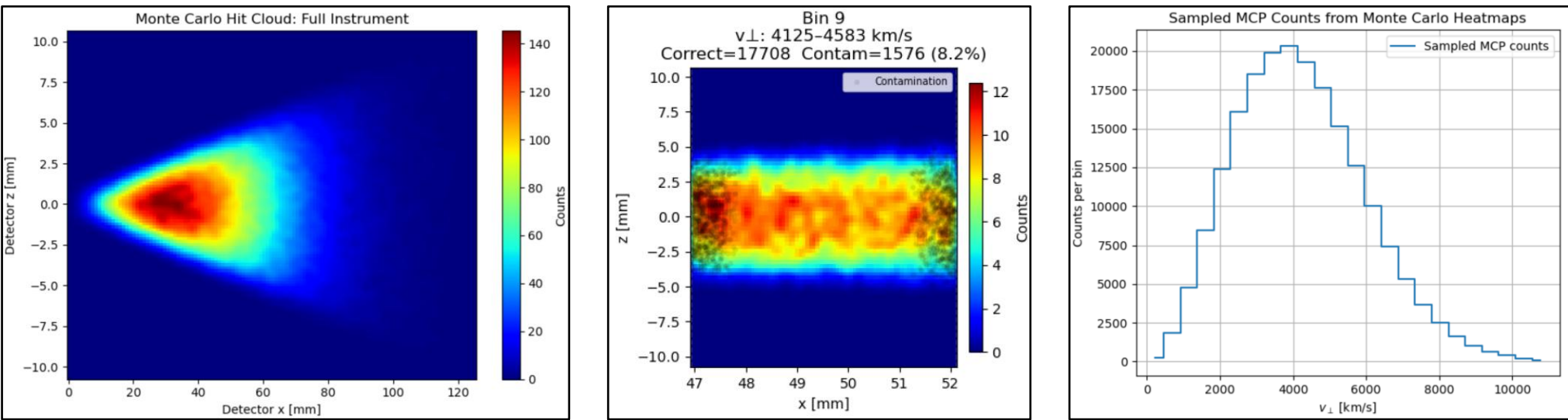
Preliminary Results + Future Work

Preliminary Results

- Multi-aperture geometry increases detector coverage.
- Electron velocity maps to detector position via magnetic deflection.
- Preliminary simulations demonstrate velocity-space discrimination.
- Monte Carlo framework enables geometry optimisation studies.



Single Aperture Results



- Distinct detector response observed for individual velocity bins.
- Electron velocity maps directly to detector impact position via magnetic deflection.
- Preliminary contamination estimates indicate velocity-space discrimination is achievable.
- **Smoothing is applied only to plotted heatmaps for visual clarity; contamination estimates use raw Monte Carlo particle counts.**

References and Acknowledgments

[1] Chen, C.H.K., Klein, K.G. & Howes, G.G. Evidence for electron Landau damping in space plasma turbulence. *Nat Commun* **10**, 740 (2019). <https://doi.org/10.1038/s41467-019-08435-3>
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Future Development

- SIMION trajectory and material modelling
- MEfit geometry and aperture optimisation
- Magnetic field homogeneity testing
- Breadboard development and detector testing
- Validate model predictions against prototype measurements